



Holes vs. Asymptotes

Factor First, Then Decide!

Ms. Goforth

Amarillo High School

Algebra 2 • 2026

* The Rule // Cancels or Stays?

• How To Tell

Step 1: Factor top and bottom completely.

Step 2: Look at each denominator factor:

Factor **cancels** (top & bottom) | → **HOLE**

Factor **stays** in denominator | → **Vertical Asymptote**

* Remember

Both holes AND asymptotes are **excluded from the domain!**

* Hole or VA?

$$f(x) = \frac{x^2 - 4}{x - 2}$$

* Hole or VA?

$$f(x) = \frac{x^2 - 4}{x - 2}$$

Factor: $\frac{(x + 2)(x - 2)}{(x - 2)}$

$(x - 2)$ appears in **both** → it **cancels**!

Hole at $x = 2$

Simplified: $f(x) = x + 2$ (with a hole at $x = 2$)

* Hole or VA?

$$f(x) = \frac{x^2 - 9}{x + 3}$$

* Hole or VA?

$$f(x) = \frac{x^2 - 9}{x + 3}$$

Factor: $\frac{(x + 3)(x - 3)}{(x + 3)}$

$(x + 3)$ cancels!

Hole at $x = -3$

Simplified: $f(x) = x - 3$

* Hole or VA?

$$f(x) = \frac{x^2 + 5x}{x}$$

* Hole or VA?

$$f(x) = \frac{x^2 + 5x}{x}$$

Factor: $\frac{x(x+5)}{x}$

The x cancels!

Hole at $x = 0$

Simplified: $f(x) = x + 5$

* Hole or VA?

$$f(x) = \frac{x+1}{x-4}$$

* Hole or VA?

$$f(x) = \frac{x+1}{x-4}$$

Already factored. Nothing cancels!

$(x - 4)$ **stays** in the denominator.

VA at $x = 4$

* Hole or VA?

$$f(x) = \frac{x^2 - 1}{x + 3}$$

* Hole or VA?

$$f(x) = \frac{x^2 - 1}{x + 3}$$

Factor top: $\frac{(x + 1)(x - 1)}{(x + 3)}$

Nothing cancels! $(x + 3)$ stays.

VA at $x = -3$

Watch Out!

Some functions have a hole AND a VA. Factor everything!

$$f(x) = \frac{x^2 + 5x + 6}{x^2 + x - 2}$$

* Hole or VA? // Both!

Watch Out!

Some functions have a hole AND a VA. Factor everything!

$$f(x) = \frac{x^2 + 5x + 6}{x^2 + x - 2}$$

Factor: $\frac{(x+2)(x+3)}{(x+2)(x-1)}$

$(x+2)$ cancels $\rightarrow$ **Hole at** $x = -2$

$(x-1)$ stays $\rightarrow$ **VA at** $x = 1$

Hole: $x = -2$

VA: $x = 1$

* Hole or VA? // Both!

$$f(x) = \frac{x^2 - x - 6}{x^2 - 9}$$

* Hole or VA? // Both!

$$f(x) = \frac{x^2 - x - 6}{x^2 - 9}$$

Factor: $\frac{(x-3)(x+2)}{(x-3)(x+3)}$

$(x-3)$ cancels $\rightarrow$ **Hole at** $x = 3$

$(x+3)$ stays $\rightarrow$ **VA at** $x = -3$

Hole: $x = 3$

VA: $x = -3$

* Hole or VA? // Both!

$$f(x) = \frac{2x^2 - 8}{x^2 - x - 2}$$

* Hole or VA? // Both!

$$f(x) = \frac{2x^2 - 8}{x^2 - x - 2}$$

Factor: $\frac{2(x-2)(x+2)}{(x-2)(x+1)}$

$(x-2)$ cancels $\rightarrow$ **Hole at** $x = 2$

$(x+1)$ stays $\rightarrow$ **VA at** $x = -1$

Hole: $x = 2$

VA: $x = -1$

* Your Turn!

▷ *YouTry!*

Factor first, then decide: Hole, VA, or both?

1. $f(x) = \frac{x^2 - 16}{x - 4}$

Hole at _____ or VA at _____

2. $f(x) = \frac{x^2 + 3x - 10}{x^2 - 25}$

Hole at _____ VA at _____

* Your Turn!

▷ YouTry!

Factor first, then decide: Hole, VA, or both?

$$1. f(x) = \frac{x^2 - 16}{x - 4}$$

Hole at _____ or VA at _____

$$2. f(x) = \frac{x^2 + 3x - 10}{x^2 - 25}$$

Hole at _____ VA at _____

$$1. (x + 4)(\cancel{x - 4}) / (\cancel{x - 4}) \rightarrow \text{Hole at } x = 4$$

$$2. (\cancel{x + 5})(x - 2) / (\cancel{x + 5})(x - 5) \rightarrow \text{Hole at } x = -5, \text{ VA at } x = 5$$



Go Sandies!



Factor first. Ask questions later.